



Lecture 4 – Threat Intelligence & Threat Hunting



Introduction to Threat Intelligence

What is a Threat?

Before understanding Threat Intelligence, we must first understand the meaning of a **Threat**.

A threat is anything that has the potential to harm an organization's:



Systems



Network



Data



Users



Business Operations

Simple Definition

A Threat is a potential danger that can cause damage if not prevented.



Real-Life Example

Suppose a ransomware attack successfully targets one bank.

Now that ransomware group may target:



Other Banks



Hospitals



Companies



Government Organizations

That ransomware campaign becomes a threat to similar organizations.

What is Threat Intelligence?

Threat Intelligence (TI) is the process of collecting, analyzing, and understanding information about cyber threats, attackers, their infrastructure, techniques, tools, and motivations.

Official Definition

Threat Intelligence is information about cyber threats that helps organizations predict, detect, prevent, and respond to cyber attacks.

Daily Life Example

Imagine a neighborhood receives information that thieves are entering houses through unlocked windows at night.

Residents can:

- ☒ Lock Windows
- ☒ Install Cameras
- ☒ Increase Security

before becoming victims.

Threat Intelligence works exactly the same way in cybersecurity.

How Threat Intelligence is Collected

Suppose a ransomware attack occurs on a bank.

SOC Analysts and Threat Researchers collect:

File Information

- Malware Sample
 - File Name
 - File Path
 - File Behavior
-

Hash Values

Calculate:

- MD5
- SHA1
- SHA256

hashes of the malware file.

These hashes are then checked using:

[VirusTotal](https://www.virustotal.com/)

to identify:

- Detection Results
 - Malware Family
 - Reputation
-

Network Information

Investigate:

- IP Addresses
- Domains
- URLs
- C2 Servers

Using reputation platforms.

OSINT (Open Source Intelligence)

What is OSINT?

OSINT stands for:

Open Source Intelligence

It refers to collecting information from publicly available sources.

Sources of OSINT

-  Blogs
 -  Google Searches
 -  News Articles
 -  Research Papers
 -  Threat Intelligence Websites
 -  Dark Web Monitoring
 -  Social Media
 -  Public Databases
-

Example

If an attacker uses:

malicious-bank-login.com

analysts can search OSINT sources to determine:

- Domain Owner
 - Hosting Provider
 - Reputation
 - Previous Attacks
-

Why Companies Use Threat Intelligence?

This is one of the most common SOC Analyst interview questions.

Interview Question

-  Why do companies invest in Threat Intelligence?
-

1 Early Warning System

Threat Intelligence provides alerts about emerging threats before they affect the company.

Example

If a phishing campaign targets banks using:

secure-bank-login.xyz

Threat Intelligence feeds can warn organizations before employees receive those emails.

2 Better Security Detection

Threat Intelligence provides:

IOCs (Indicators of Compromise)

Examples:

- ✓ Malicious IPs
- ✓ Domains
- ✓ URLs
- ✓ File Hashes

These can be added to:

- SIEM
- EDR
- IDS
- IPS
- Firewalls

for automatic detection.

3 Understanding Threat Actors

Threat Intelligence helps answer:

Who is attacking?

Why are they attacking?

What do they want?

Example

A ransomware group specifically targeting banks for financial gain.

4 Reducing Alert Fatigue

SOC teams receive thousands of alerts daily.

Threat Intelligence helps prioritize:

 Real Threats

vs

 False Positives

This saves analyst time.

5 Faster Incident Response

During investigations, Threat Intelligence provides context.

Example

Analyst discovers:

SHA256: ABC123...

Threat Intelligence identifies it as:

 Emotet Malware

or

 TrickBot Malware

This accelerates response.

Compliance & Risk Management

Many security frameworks require proactive monitoring.

Examples:

- NIST
- International Organization for Standardization (ISO 27001)
- GDPR

Threat Intelligence demonstrates proactive security practices.

Types of Threat Intelligence

Threat Intelligence is divided into four major categories.

Strategic Threat Intelligence

Audience

-  Executives
 -  Management
 -  CISOs
-

Purpose

Provides a high-level business perspective.

Answers:

- Who is attacking?
 - Why are they attacking?
 - What business risks exist?
-

Information Included

- Threat Actor Groups
 - Motivations
 - Targeted Industries
 - Regional Threat Trends
-

Example

"A ransomware group is increasingly targeting healthcare organizations across Asia."

SOC Benefit

Helps justify:

-  Security Budgets
 -  Security Policies
 -  Security Investments
-

2 Tactical Threat Intelligence

Audience

-  SOC Analysts
 -  Security Engineers
-

Purpose

Explains how attackers operate.

Information Included

- Attack Techniques
 - Malware Families
 - Tools Used
 - Attack Patterns
-

Examples

- Phishing
- Privilege Escalation
- Credential Theft

Framework commonly used:

MITRE ATT&CK

Malware Examples

- Mimikatz
 - Cobalt Strike
 - Emotet
-

SOC Benefit

Used to improve:

- SIEM Rules
 - IDS Signatures
 - Detection Logic
-

3 Operational Threat Intelligence

Audience

 Incident Response Teams

 Threat Hunters

Purpose

Provides information about ongoing attacks.

Information Included

- Current Attack Activity
 - TTPs
 - Attack Timelines
 - C2 Infrastructure
-

Example

Current attack uses:

- PowerShell
 - External C2 Server
 - Data Exfiltration
-

SOC Benefit

Helps teams predict attacker behavior.

Technical Threat Intelligence

Audience

Security Tools

- SIEM
 - EDR
 - Firewalls
 - IDS/IPS
-

Purpose

Provides machine-readable threat data.

Examples of IOCs

 Malicious IP Addresses

 Domains

 URLs

 Email Addresses

 Hash Values

Example

Block:

192.168.10.45

Hash:

ABC123XYZ

Domain:

badmail.xyz

SOC Benefit

Automated blocking and detection.

Easy Interview Shortcut

Type	Focus
Strategic	Who & Why
Tactical	How
Operational	What is Happening Now
Technical	Exact IOCs

Free Threat Intelligence Tools

AlienVault OTX

[AlienVault OTX](#)

Provides:

- Threat Pulses
 - IOCs
 - Malware Information
 - Threat Sharing
-

MISP

[MISP Project](#)

Open-source Threat Intelligence sharing platform.

Useful for:

- IOC Sharing

- SOC Labs
 - Threat Collaboration
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Abuse.ch Projects

[Abuse.ch](https://abuse.ch)

Provides:

- URLhaus
- Feodo Tracker
- MalwareBazaar

Useful for malware and botnet intelligence.

VirusTotal

[VirusTotal](https://www.virustotal.com)

Used for:

- File Analysis
 - URL Analysis
 - Reputation Checking
-

OSINT Investigation Tools

Shodan

[Shodan](https://shodan.io)

Search engine for internet-connected devices.

Used to find:

- Open Ports
- Exposed Servers
- Vulnerable Services

Censys

[Censys](#)

Similar to Shodan.

Provides internet asset visibility.

Maltego

[Maltego Community Edition](#)

Visual OSINT investigation platform.

Used to map:

- Domains
 - People
 - Infrastructure
 - Threat Actors
-

Whois / Google Dorks / Hunter.io

Used for:

- Domain Intelligence
 - Email Intelligence
 - Organization Research
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Malware Analysis Platforms

Hybrid Analysis

[Hybrid Analysis](#)

Online malware sandbox.

Any.Run

[ANY.RUN](#)

Interactive malware analysis sandbox.

Popular among SOC Analysts.

Cuckoo Sandbox

[Cuckoo Sandbox](#)

Open-source malware analysis environment.



Threat Intelligence Feeds

STIX / TAXII

Industry standards for threat intelligence sharing.

Used by SIEM platforms.

Spamhaus

[Spamhaus](#)

Provides:

- Spam Domains
 - Malicious IP Lists
-

PhishTank

[PhishTank](#)

Database of phishing URLs.

Cisco Talos

[Cisco Talos Intelligence Group](#)

Provides:

- Threat Reports
 - Reputation Services
 - Malware Intelligence
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Threat Intelligence Lifecycle in a Company

1 Collection

Threat data comes from:

External Sources

- OTX
- MISP
- OSINT
- Security Vendors

Internal Sources

- SIEM Logs
- Firewall Logs
- EDR Alerts
- IDS/IPS

Dark Web Monitoring

Monitoring:

- Stolen Credentials
 - Leaked Data
-

2 Processing

Raw data is cleaned and enriched.

Examples:

- ✓ Remove Duplicates
 - ✓ Add WHOIS Information
 - ✓ Add Reputation Scores
 - ✓ Add Geolocation
-

3 Analysis

Security teams analyze:

IOCs

- IPs
- Domains
- Hashes

TTPs

Tactics, Techniques, and Procedures.

Attribution

Who is behind the attack?

Example:

FIN7

APT29

Ransomware Groups

Dissemination

Intelligence is shared with:

-  SOC Team
 -  Incident Response Team
 -  Management
-

5 Action

Actions include:

-  Block IPs
 -  Block Domains
 -  Update SIEM Rules
 -  Update EDR Rules
 -  Patch Vulnerabilities
 -  User Awareness Training
-

6 Feedback

Continuous improvement cycle.

Collect → Process → Analyze → Share → Act → Repeat

Threat vs Vulnerability vs Risk

Threat

Potential danger.

Example

Attacker sending phishing email.

Vulnerability

Weakness that can be exploited.

Examples

- ✗ Weak Password
 - ✗ Outdated Software
 - ✗ Misconfigured Firewall
-

Risk

Probability that a threat will exploit a vulnerability.

Formula

$$Risk = Threat \times Vulnerability \times Impact$$

Example

Threat = Hacker

Vulnerability = Password123

Risk = High chance of compromise

Easy Analogy

 Threat = Thief

 Vulnerability = Unlocked Door

 Risk = Chance of Being Robbed

What is Threat Hunting?

Threat Hunting is a proactive cybersecurity activity where analysts actively search for hidden threats that traditional security tools may have missed.

Simple Definition

Threat Hunting means searching for attackers before they cause damage.

Difference Between SOC Monitoring and Threat Hunting

SOC Monitoring Threat Hunting

Waits for alerts	Searches proactively
Reactive	Proactive
Tool Driven	Analyst Driven

Why Threat Hunting is Needed

Because:

-  Antivirus can be bypassed
 -  Malware can hide
 -  Fileless attacks exist
 -  Insider threats exist
 -  Zero-Day attacks have no signatures
-

What Threat Hunters Look For

-  Unusual Logins
-  Suspicious PowerShell
-  Malicious IP Connections

-  Persistence Mechanisms
 -  Lateral Movement
 -  Data Exfiltration
 -  Abnormal Network Traffic
-

Threat Hunting Process

1. Hypothesis

Example:

"Has anyone executed suspicious PowerShell commands in the last 7 days?"

2. Data Collection

Sources:

- SIEM
 - EDR
 - Firewall Logs
 - DNS Logs
 - Endpoint Logs
-

3. Investigation

Analyze:

- IOCs
 - TTPs
 - MITRE ATT&CK Mapping
-

4. Detection

If malicious activity is found:

- ✓ Create IOC
 - ✓ Write Detection Rule
 - ✓ Block Attacker
-

5. Response

Actions:

- ✗ Disable User
 - ✗ Block IP
 - ✗ Quarantine Endpoint
 - ✓ Patch Vulnerability
-

Goal of Threat Hunting

- ✓ Reduce Dwell Time
 - ✓ Detect Hidden Threats
 - ✓ Improve SOC Visibility
 - ✓ Create Better Detection Rules
 - ✓ Strengthen Security
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Popular Threat Hunting Tools

SIEM

- Splunk
- IBM QRadar
- Microsoft Sentinel

EDR

- CrowdStrike

- SentinelOne

Other Tools

- Wazuh
- ELK Stack
- Wireshark
- Zeek
- VirusTotal
- OTX
- Shodan

Understanding the Threat Landscape

Threat Actors (Who?)

Cyber Criminals

Goal:

Money

Examples:

- Ransomware
- Banking Fraud

Hacktivists

Goal:

Political or Social Causes

Example:

Website Defacement

Nation-State Actors

Goal:

Cyber Warfare

Espionage

Military Intelligence

Insider Threats

Goal:

Revenge

Greed

Negligence

Script Kiddies

Goal:

Fun

Learning

Recognition

Use publicly available hacking tools.

Easy Interview Answer

Threat

Potential danger.

Threat Actor

The person or group causing the threat.

Motive

The reason why they are attacking.

Most Important Interview Questions

Q1. What is Threat Intelligence?

Threat Intelligence is the collection and analysis of information about cyber threats to help organizations detect, prevent, and respond to attacks.

Q2. Why is Threat Intelligence important?

It provides early warning, improves detection, reduces false positives, supports incident response, and strengthens overall security.

Q3. What are the four types of Threat Intelligence?

- ✓ Strategic
- ✓ Tactical
- ✓ Operational
- ✓ Technical

Q4. What is Threat Hunting?

Threat Hunting is a proactive process of searching for hidden threats that may have bypassed security tools.

Q5. Difference between Threat and Threat Actor?

Threat = Potential danger.

Threat Actor = Person or group causing that danger.
